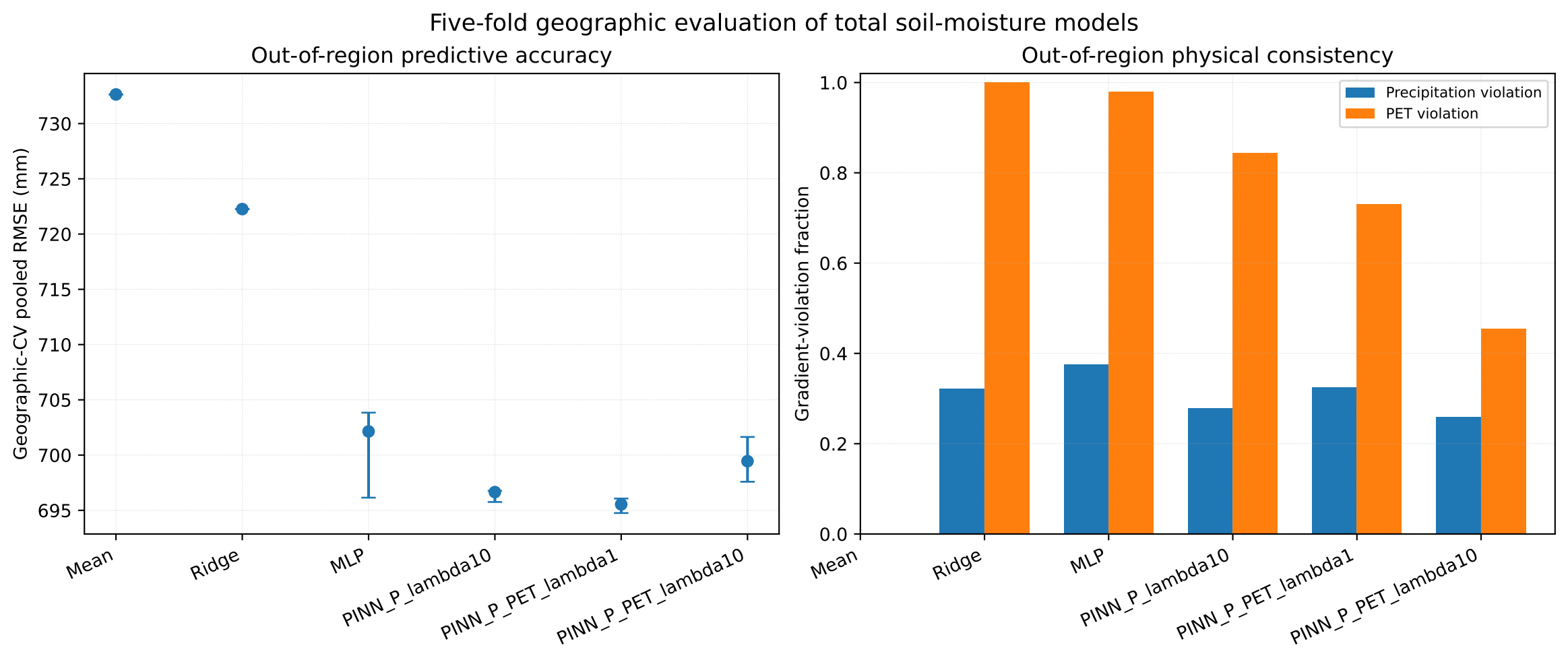
Five-fold geographic evaluation of total soil-moisture models
Out-of-region predictive accuracy
Geographic-CV pooled RMSE (mm)
730
725
720
715
710
705
700
695
Mean
Ridge
MLP
PINN_P_lambda10
PINN_P_PET_lambda1
PINN_P_PET_lambda10
Out-of-region physical consistency
Gradient-violation fraction
1.0
0.8
0.6
0.4
0.2
0.0
Precipitation violation
PET violation
Mean
Ridge
MLP
PINN_P_lambda10
PINN_P_PET_lambda1
PINN_P_PET_lambda10